Introductory Calculus

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The derivative as a limit**

Write the difference-quotient definition of $f'(x)$.

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

2 **[Foundational]** • **Target: The derivative as a limit**

Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.

$f'(3) = 2(3) = 6$.

3 **[Foundational]** • **Target: Differential equations**

Explain why solving a differential equation gives a family of functions rather than a single number.

A differential equation only constrains the derivative, not the exact function; any constant added to a valid solution y has derivative 0 and doesn't change whether it solves the equation, so every $y =$ (that solution) $+C$ also works — a whole family, not one answer.

4 **[Foundational]** • **Target: Differential equations**

Evaluate the indeterminate limit: $\lim_{x \to 5} \frac{x^2 - 25}{x^2 - 8x + 15}$

Step 1: Direct substitution check: $\frac{5^2 - 25}{5^2 - 40 + 15} = \frac{0}{0}$ (indeterminate). Step 2: Factor numerator and denominator: $\frac{(x-5)(x+5)}{(x-5)(x-3)}$ Step 3: Cancel common factor and evaluate: $\lim_{x \to 5} \frac{x+5}{x-3} = \frac{5+5}{5-3} = \frac{10}{2} = \mathbf{5}$.

5 **[Foundational]** • **Target: Direct substitution**

Evaluate the limit at infinity involving a square root in the numerator: $\lim_{x \to \infty} \frac{\sqrt{9x^2 + 4x}}{2x - 1}$

Step 1: Divide numerator and denominator by x (for $x > 0$, $x = \sqrt{x^2}$): $\frac{\sqrt{9x^2 + 4x}}{2x - 1} = \frac{\frac{\sqrt{9x^2 + 4x}}{\sqrt{x^2}}}{\frac{2x-1}{x}} = \frac{\sqrt{\frac{9x^2}{x^2} + \frac{4x}{x^2}}}{2 - \frac{1}{x}} = \frac{\sqrt{9 + \frac{4}{x}}}{2 - \frac{1}{x}}$. Step 2: Take the limit as $x \to \infty$: Since $\frac{4}{x} \to 0$ and $\frac{1}{x} \to 0$.

6 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{\tan(2x)}$

Step 1: Express $\tan(2x)$ in terms of sine and cosine: $\frac{\sin(5x)}{\tan(2x)} = \frac{\sin(5x)}{\frac{\sin(2x)}{\cos(2x)}} = \frac{\sin(5x)}{\sin(2x)} \cdot \cos(2x)$ Step 2: Multiply and divide by appropriate factors of x: $\frac{\sin(5x)}{5x} \cdot 5x \cdot \frac{2x}{\sin(2x)} \cdot \frac{1}{2x} \cdot \cos(2x) = \frac{\sin(5x)}{5x} \cdot \frac{1}{\frac{\sin(2x)}{2x}} \cdot \frac{5}{2} \cdot \cos(2x)$ Step 3: Evaluate limits using $\lim_{u \to 0} \frac{\sin u}{u} = 1$ and $\cos(0) = 1$.

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7 [Foundational] • Target: The derivative as a limit

Evaluate the limit using algebraic conjugate rationalization: $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$

Step 1: Check form upon direct substitution: $\frac{\sqrt{4}-2}{4-4} = \frac{0}{0}$ (indeterminate form). Step 2: Multiply by the conjugate of the numerator: $\lim_{x\to 4} \frac{(\sqrt{x}-2)(\sqrt{x}+2)}{(x-4)(\sqrt{x}+2)} = \lim_{x\to 4} \frac{(\sqrt{x})^2-2^2}{(x-4)(\sqrt{x}+2)} = \lim_{x\to 4} \frac{x-4}{(x-4)(\sqrt{x}+2)}$. Step 3: Cancel the common factor $(x-4)$ for $x \neq 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Differential equations

Evaluate the limit of the rational expression by factoring: $\lim_{x\to 3} \frac{x^3-27}{x^2-2x-3}$

Step 1: Direct substitution check: $\frac{3^3-27}{3^2-2(3)-3} = \frac{27-27}{9-6-3} = \frac{0}{0}$ (indeterminate). Step 2: Factor numerator using difference of cubes: $x^3-27 = (x-3)(x^2+3x+9)$ Step 3: Factor denominator: $x^2-2x-3 = (x-3)(x+1)$ Step 4: Cancel the common factor $(x-3)$ and evaluate.

9 [Intermediate] • Target: Direct substitution

Solve the nonlinear system of equations: $\{y = x^2-4x+3;\ 2x+y=6$ Find all points of intersection (x, y).

Step 1: Isolate y from the linear equation: $y = 6-2x$ Step 2: Substitute into the quadratic equation: $6-2x = x^2-4x+3 \Rightarrow x^2-2x-3=0$. Step 3: Factor the quadratic: $(x-3)(x+1)=0 \Rightarrow x_1=3, \quad x_2=-1$. Step 4: Compute corresponding y-values:For $x=3$: $y=6-2(3)=0 \Rightarrow \mathbf{(3,0)}$.For $x=-1$: $y=6-2(-1)=8 \Rightarrow \mathbf{(-1,8)}$. The points of intersection are $\mathbf{(3,0)}$ and $\mathbf{(-1,8)}$.

10 [Intermediate] • Target: Indeterminate forms: factor and cancel

Evaluate the limit at infinity: $\lim_{x\to\infty} \frac{5x^3-2x+7}{2x^3+4x^2-1}$

Step 1: Compare degrees of numerator and denominator: Both numerator and denominator have degree 3. Step 2: Apply Theorem 6.1 (Ratio of leading coefficients): $\lim_{x\to\infty} \frac{5x^3-2x+7}{2x^3+4x^2-1} = \frac{5}{2} = \mathbf{\frac{5}{2}}$.

11 [Intermediate] • Target: The derivative as a limit

In ΔABC, $a=7$, $b=9$, and $c=10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a=7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{9^2+10^2-7^2}{2(9)(10)} = \frac{81+100-49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7+9+10}{2} = 13$..

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12 [Intermediate] • Target: Differential equations

Find the modulus and principal argument of $z = -2\sqrt{3} + 2i$, and express z in polar form $r(\cos\theta + i\sin\theta)$.

Step 1: Compute modulus r: $r = \sqrt{(-2\sqrt{3})^2 + 2^2} = \sqrt{12+4} = \sqrt{16} = 4$. Step 2: Determine argument θ: Since $a = -2\sqrt{3} < 0$ and $b = 2 > 0$, z lies in Quadrant II. Reference angle: $\tan\alpha = \left|\frac{2}{-2\sqrt{3}}\right| = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$ (or 30°). Quadrant II angle: $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ (or 150°). Step 3: Write polar form.

13 [Intermediate] • Target: Direct substitution

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

14 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find the distance in the polar coordinate plane between the two points $P_1\left(4, \frac{\pi}{3}\right)$ and $P_2\left(7, \frac{2\pi}{3}\right)$.

Step 1: Apply the Law of Cosines for polar distance: $d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_2 - \theta_1)}$ Step 2: Substitute values: $r_1 = 4$, $r_2 = 7$, $\theta_2 - \theta_1 = \frac{2\pi}{3} - \frac{\pi}{3} = \frac{\pi}{3}$. $d = \sqrt{4^2 + 7^2 - 2(4)(7)\cos\left(\frac{\pi}{3}\right)} = \sqrt{16 + 49 - 56\left(\frac{1}{2}\right)}$ $d = \sqrt{65 - 28} = \sqrt{\mathbf{37}}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 4.01 Matrices & Vectors

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).

Step 1: Apply coordinate conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Step 2: Compute y: $y = r\sin\theta = 6\sin\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Thus the rectangular coordinates are $\mathbf{\left(-3\sqrt{2}, -3\sqrt{2}\right)}$.

16 [Challenging] • Target: Differential equations • Cross-Lesson: 6.01 Probability

Convert the Cartesian equation $x^2 + (y-4)^2 = 16$ into a polar equation relating r and θ.

Step 1: Expand the binomial: $x^2 + y^2 - 8y + 16 = 16 \Rightarrow x^2 + y^2 - 8y = 0$. Step 2: Substitute polar identities $x^2 + y^2 = r^2$ and $y = r\sin\theta$: $r^2 - 8r\sin\theta = 0 \Rightarrow r(r - 8\sin\theta) = 0$. Step 3: Divide by r for $req0$ (the origin is included at $\theta = 0$): $\mathbf{r = 8\sin\theta}$.

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17 [Challenging] • Target: Direct substitution • Cross-Lesson: 1.01 Complex Numbers

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

18 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3,\ -3\sqrt{3})$.

19 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x-4)^2 + (y+3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4,\ -3)$ and radius $r = \sqrt{36} = 6$.

20 [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Find the exact value of $\sin 30^\circ + \cos 60^\circ$.

Step 1: Recall exact unit circle trigonometric ratios:$\sin 30^\circ = \frac{1}{2}$.$\cos 60^\circ = \frac{1}{2}$ (by cofunction identity $\cos(90^\circ - \theta) = \sin\theta$).Step 2: Add values: $\frac{1}{2} + \frac{1}{2} = \mathbf{1}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct substitution

Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

a. 1 ✓

b. -1

c. 3

d. 4

$f'(x) = -2x + 5$, so $f'(2) = -2(2) + 5 = -4 + 5 = 1$.

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22 [Foundational] • Target: Indeterminate forms: factor and cancel

Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 0$.

a. -3

b. 3

c. 1

d. -1 ✓

$\frac{f(0)-f(-1)}{0-(-1)} = \frac{0^2-(-1)^2}{1} = \frac{-1}{1} = -1$.

23 [Foundational] • Target: The derivative as a limit

If $f(x) = 7x - 4$, find $f'(x)$ using the difference-quotient definition.

a. **7** ✓

b. -4

c. 8

d. 0

$\frac{f(x+h)-f(x)}{h} = \frac{7(x+h)-4-(7x-4)}{h} = \frac{7h}{h} = 7$, so $f'(x) = 7$ for every x (constant slope).

24 [Foundational] • Target: Differential equations

How many ways can 6 items be arranged, taken 2 at a time (order matters)?

a. **30** ✓

b. 15

c. 32

d. 720

Applying Probability principles, $P(6, 2) = \frac{6!}{(6-2)!} = 30$. Therefore, the verified correct option is 30.

25 [Foundational] • Target: Direct substitution

A fair coin is flipped 3 times. Find $P(\text{all heads})$.

a. $\frac{1}{8}$ ✓

b. $\frac{1}{6}$

c. $\frac{3}{8}$

d. $\frac{1}{4}$

Each flip is independent with $P(\text{H}) = \frac{1}{2}$: $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

26 [Foundational] • Target: Indeterminate forms: factor and cancel

Independent events A, B have $P(A) = \frac{4}{9}$ and $P(B) = \frac{1}{3}$. Find $P(A \text{ and } B)$.

a. $\frac{4}{27}$ ✓

b. $\frac{7}{9}$

c. $\frac{6}{41}$

d. $\frac{7}{18}$

For independent events, $P(A \text{ and } B) = P(A)P(B) = \frac{4}{9} \times \frac{1}{3} = \frac{4}{27}$.

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27 [Foundational] • Target: The derivative as a limit

A fair six-sided die is rolled. Find $P(\text{rolling a } 3)$.

a. $\frac{1}{6}$ ✓ **b.** $\frac{1}{3}$

c. $\frac{1}{2}$ **d.** $\frac{5}{6}$

One favorable outcome out of 6 equally likely outcomes: $P = \frac{1}{6}$.

28 [Foundational] • Target: Differential equations

A committee of 3 men and 1 women is to be chosen from 6 men and 5 women. How many ways?

a. **100** ✓ **b.** 25

c. 330 **d.** 106

Choose the men and women independently, then multiply: $6351 = 205 = 100$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Direct substitution

A card is drawn from a standard 52-card deck. Find $P(\text{face card (J, Q, or K)})$.

a. $\frac{3}{13}$ ✓ **b.** $\frac{3}{52}$

c. $\frac{1}{13}$ **d.** $\frac{1}{4}$

12 face cards out of 52: $P = \frac{12}{52} = \frac{3}{13}$.

30 [Intermediate] • Target: Indeterminate forms: factor and cancel

A bag has 5 black marbles and 7 white marbles. One is drawn at random. Find $P(\text{black})$.

a. $\frac{5}{12}$ ✓ **b.** $\frac{7}{12}$

c. $\frac{5}{7}$ **d.** $\frac{1}{2}$

$P(\text{black}) = \frac{5}{12} = \frac{5}{12}$.

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31 [Intermediate] • Target: The derivative as a limit

A bag has 5 green marbles and 2 yellow marbles. One is drawn at random. Find $P(\text{green})$.

a. $\frac{5}{7}$ ✓ **b.** $\frac{2}{7}$

c. $\frac{5}{2}$ **d.** $\frac{1}{2}$

$P(\text{green}) = \frac{5}{7} = \frac{5}{7}$.

32 [Intermediate] • Target: Differential equations

A bag has 2 red marbles and 7 blue marbles. One is drawn at random. Find $P(\text{red})$.

a. $\frac{2}{9}$ ✓ **b.** $\frac{7}{9}$

c. $\frac{2}{7}$ **d.** $\frac{1}{2}$

$P(\text{red}) = \frac{2}{9} = \frac{2}{9}$.

33 [Intermediate] • Target: Direct substitution

How many ways can a group of 2 be chosen from 4 people (order doesn't matter)?

a. **6** ✓ **b.** 12

c. 8 **d.** 4

$42 = 4!\,2!\,2!\, = 6$.

34 [Intermediate] • Target: Indeterminate forms: factor and cancel

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

a. **4** ✓ **b.** 3

c. 1 **d.** 6

To find the roots of $P(x) = 0$, we first determine the unknown parameters a and b using the Factor Theorem and the Remainder Theorem. 1. **Apply the Factor Theorem:** Since $x - 1$ is a factor of $P(x)$, we have $P(1) = 0$: $P(1) = (1)^3 + a(1)^2 + b(1) + 12 = 0 \Rightarrow 1 + a + b + 12 = 0 \Rightarrow a + b = -13$ 2. **Apply the Remainder Theorem:** When $P(x)$ is divided by $x - 2$, the remainder is -10, so $P(2) = -10$.

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35 [Intermediate] • Target: The derivative as a limit

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. **2** ✓

b. 8

c. 18

d. -30

According to the Factor Theorem and the Remainder Theorem, if a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$. If $(x - c)$ is a factor, then $P(c) = 0$. Step 1: Set up the first equation using the factor $(x - 1)$: $P(1) = 2(1)^3 + a(1)^2 - 5(1) + b = 0$ $2 + a - 5 + b = 0 \Rightarrow a + b = 3$ Step 2: Set up the second equation using the divisor $(x + 2) = (x - (-2))$ with remainder -18: $P(-2) = 2(-2)^3 + a(-2)^2 - 5(-2) + b = -18$ $2(-8) + 4a + 10 + b = -18$.

36 [Intermediate] • Target: Differential equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 - x + 4$.

a. $4x^2 - 1$

b. $8x + 4$

c. $4x - 1$

d. $8x - 1$ ✓

Expand $f(x + h) - f(x) = 4(2xh + h^2) - 1h = (8x - 1)h + 4h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 8x - 1 + 4h$. Letting $h \to 0$: $f'(x) = 8x - 1$.

37 [Intermediate] • Target: Direct substitution

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Sequences & Series

What is the minimum value of the function $f(x) = \sqrt{x^2 - 4x + 13} + \sqrt{x^2 - 14x + 130}$ for all real numbers x?

a. 12

b. $4\sqrt{10}$

c. **13** ✓

d. $\sqrt{170}$

To find the minimum value of $f(x)$, synthesize the algebraic expressions with coordinate geometry by completing the square under each radical: 1. Rewrite the radicands: $x^2 - 4x + 13 = (x - 2)^2 + 9 = (x - 2)^2 + 3^2$ $x^2 - 14x + 130 = (x - 7)^2 + 81 = (x - 7)^2 + 9^2$ 2. Interpret geometrically using the Euclidean distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$: - $\sqrt{(x - 2)^2 + 3^2}$ is the distance from the point $P(x, 0)$ on the x-axis to the fixed point $A(2, 3)$..

39 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 3.01 Functions & Graphs

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. **-2** ✓

b. 2

c. 7

d. 16

Let the equation of the tangent line L be $y = mx + b$. Because L is tangent to the quartic polynomial $f(x)$ at two distinct points, say $x = r$ and $x = s$ with res, each point of tangency corresponds to a root of multiplicity at least 2 for the difference equation $f(x) - (mx + b) = 0$. Since $f(x) - (mx + b)$ is a monic polynomial of degree 4, it factors completely over the real numbers as: $f(x) - (mx + b) = (x - r)^2(x - s)^2$ Expanding this product.

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40 [Challenging] • Target: Differential equations • Cross-Lesson: 4.01 Matrices & Vectors

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

41 [Challenging] • Target: Direct substitution • Cross-Lesson: 6.01 Probability

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$ ✓

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

We are given that the point $(-2, 5)$ lies on the graph of $y = f(x)$, which means $f(-2) = 5$. The function g is defined as $g(x) = f(x - 3) - 4$. To evaluate $g(x)$ using our known value $f(-2)$, we set the argument of f equal to -2: $x - 3 = -2 \Rightarrow x = 1$. Now, substitute $x = 1$ into the expression for $g(x)$: $g(1) = f(1 - 3) - 4 = f(-2) - 4 = 5 - 4 = 1$. Therefore, the point $(1, 1)$ must lie on the graph of $y = g(x)$. Geometrically, the transformation $y = f(x - 3) - 4$ translates the graph of.

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42 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 1.01 Complex Numbers

If the limit $\lim_{x\to 3}\frac{x^2+kx-15}{x^2-4x+3}=L$ exists and is a finite real number, where k and L are constants, what is the value of $k+L$?

a. 2

b. 4

c. **6** ✓

d. 8

First, evaluate the limit of the denominator as $x\to 3$: $3^2-4(3)+3=9-12+3=0$ Since the denominator approaches 0 and the limit L is a finite real number, the numerator must also approach 0 as $x\to 3$ to produce an indeterminate form of $\frac{0}{0}$. Setting the numerator equal to 0 at $x=3$: $3^2+3k-15=0\Rightarrow 9+3k-15=0\Rightarrow 3k-6=0\Rightarrow k=2$ Next, substitute $k=2$ back into the rational function and factor both numerator and denomin.

43 [Challenging] • Target: The derivative as a limit • Cross-Lesson: 2.01 Sequences & Series

For each real number $t\in(-2,2)$, the cubic equation $x^3-3x+t=0$ has three distinct real roots, denoted in increasing order by $x_1(t)<x_2(t)<x_3(t)$. What is the value of the limit $\lim_{t\to 2^-}\frac{x_3(t)-x_2(t)}{\sqrt{2-t}}$?

a. $\frac{2\sqrt{3}}{3}$ ✓

b. $\sqrt{3}$

c. $\frac{\sqrt{3}}{2}$

d. $\frac{4}{3}$

To evaluate the limit, we analyze the behavior of the roots as $t\to 2^-$ using polynomial factorization and Vieta's formulas. **Step 1: Determine the limiting values of the roots.** When $t=2$, the equation becomes: $x^3-3x+2=(x-1)^2(x+2)=0$ The roots are -2 (simple root) and 1 (double root). Since $x_1(t)x_2(t)$, the difference $x_3(t)-x_2(t)$ is strictly positive. Taking the square root gives.

44 [Challenging] • Target: Differential equations • Cross-Lesson: 3.01 Functions & Graphs

What is the value of the limit $\lim_{x\to 2}\frac{x^2+x-6}{x-2}$?

a. 5 ✓

b. 1

c. -1

d. Does not exist

Direct substitution of $x=2$ gives the indeterminate form: $\frac{2^2+2-6}{2-2}=\frac{0}{0}$ To evaluate the limit, factor the quadratic expression in the numerator: $x^2+x-6=(x-2)(x+3)$ For all $xeq2$, we can cancel the common factor $(x-2)$: $\frac{x^2+x-6}{x-2}=\frac{(x-2)(x+3)}{x-2}=x+3$ Now, evaluate the limit by direct substitution: $\lim_{x\to 2}(x+3)=2+3=5$ Common misconceptions include: - Factoring incorrectly as $(x-2)(x-3)$, leading to.

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45 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Matrices & Vectors

The function f is defined by $f(x) = \frac{3x+2}{x-4}$ for all real numbers $xeq4$. What is the value of $f^{-1}(2)$?

a. -10 ✓

b. -6

c. -4

d. 6

By definition of the inverse function, finding $k = f^{-1}(2)$ is equivalent to solving $f(k) = 2$. Set up the equation: $\frac{3k+2}{k-4} = 2$ Multiply both sides by $k-4$ (where $keq4$): $3k+2 = 2(k-4)$ $3k+2 = 2k-8$ Subtract $2k$ from both sides: $k+2 = -8$ Subtract 2 from both sides: $k = -10$ Thus, $f^{-1}(2) = -10$. Common misconceptions: - Evaluating $f(2) = \frac{3(2)+2}{2-4} = \frac{8}{-2} = -4$ confuses the function f with its inverse f^{-1}. - A sign error when solving.

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